IIMBx PM Dashboard - Project Documentation & Feature Specification

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1. Executive Summary

The **IIMBx PM Dashboard** is a high-performance, AI-integrated project management tool built as an intelligence layer on top of the Freedcamp API. Designed specifically for project managers (PMs), it provides high-level visibility, automated risk detection, and a personalized AI co-manager experience to streamline decision-making and project tracking.

2. System Architecture & Technology Stack

The platform utilizes a hybrid PHP/JavaScript architecture. It is strictly optimized for deployment on shared hosting environments (such as cPanel) while delivering a modern, "App-like" performance and seamless user experience.

Layer	Technology	Description
Frontend	Vanilla HTML5, CSS3, Async JavaScript	Employs a modern Glassmorphism design system for premium aesthetics.
Backend	PHP 8.1+	Serves as an API Proxy (`api.php`) and multi-threaded cache layer to hide secrets and manage requests.

Layer	Technology	Description
Data Source	Freedcamp API (REST)	Central data repository for all project, task, and milestone tracking.
AI Engine (Analysis)	Gemini 3.1 Pro Preview	Powers deep PM Insights, automated risk audits, and complex data analysis.
AI Engine (Chat)	Gemini 3.1 Flash Lite / Flash	Drives the high-speed, real-time AI Copilot chat interface.

3. Core Capabilities & AI Integrations

3.1. AI-Powered Project Intelligence

- **Senior PM Insights Agent:** Automatically audits all project data (tasks, milestones) across the portfolio or a specific project. Generates four concise "PM Insights" (Risks, Successes, Warnings) complete with short-term workarounds and long-term strategic actions.
- **SMART Milestone Validator:** Conducts real-time evaluations of project milestones to ensure they strictly meet SMART criteria (Specific, Measurable, Achievable, Relevant, Time-bound). Automatically flags poorly defined milestones and suggests concrete improvements.
- **AI PM Copilot Chat:** A dedicated, highly-specialized chat assistant that maintains the "full story" and historical context of specific projects or the broader portfolio.
- **Multimodal Vision & Document Support:**
    - **Visual Analysis:** PMs can upload screenshots, diagrams, or site photos directly into the chat or insights bar. The multimodal Gemini engine interprets these images alongside project metadata.
    - **Document Context Extraction:** Natively supports `.txt`, `.csv`, `.pdf`, and `.docx` file uploads. The AI extracts content to inform its analysis and provide highly contextual answers.
- **Voice-to-Text Command:** Built-in microphone support enabling hands-free queries and commands.

3.2. Unified PM Interface & Dashboard

- **Project Cards:** A dynamic landing view displaying all active projects with real-time health indicators (Active/Completed).

- **Real-time Metrics & Countdown:** Automatically aggregates task counts, milestones, and overdue items. Features a custom column displaying "Days to Due Date" for every task, color-coded for urgency.
- **Personal PM Scratchpad:** A project-specific private note area designed for logging daily updates and preparing "defense notes" for project reviews.

4. User Interface & Experience (UI/UX)

The interface is designed with premium aesthetics and functional responsiveness in mind:

- **Glassmorphism Design:** Utilizes blurred backdrops, semi-transparent glass panels, and vibrant accent colors (Accent Blue: `#3498db`) coupled with subtle micro-animations.
- **Vertical Auto-Collapse:** A smart responsive layout that removes static height restrictions. The container adapts seamlessly to content volume, allowing information to flow naturally without wasted screen space.
- **Modern Iconography & Typography:** Standard browser symbols, bullets, and star-ratings have been replaced with a curated SVG icon set and clean typography (Inter/Roboto) to establish a better visual hierarchy.
- **Shimmer Loading States:** High-end skeleton animations are deployed during data fetching to maintain a smooth perception of performance.

5. Security, Authentication & Personalization

- **Secure Session Management:** Login flow and session-locked API access are strictly managed via `auth.php`.
- **Custom Name Mapping:** Specialized resolution logic detects logged-in professional emails and maps them to core team members (e.g., Harsh, Diljith, Nidhi).
- **AI Personality Layer:** The AI leverages this authentication data to power a dynamic greeting system, addressing the user by their first name across all interfaces (e.g., *"Hi Harsh! I've reviewed your latest data..."*).

6. Performance Optimization & Backend Logic

Centralized in a robust switch-case engine within `api.php`, the backend ensures stability, speed, and security:

- **4-Hour Smart Caching (Rate-Limit Defense):** A server-side file-based caching system (`ai_insights_cache_*.json`) stores AI insights. This prevents hitting Gemini API rate limits and keeps the UI snappy while ensuring data remains fresh.
- **Freedcamp Proxy & Preflight Handling:** The central API proxy masks all API keys, prevents client-side CORS issues, and features robust OPTIONS preflight handling for stable frontend-AI communication.

- **Dual-Model Orchestration:** Intelligently switches between the heavy-duty Gemini 3.1 Pro model for deep analysis and the lighter Gemini 3.1 Flash model for conversational speed.
- **Advanced Prompt Engineering:** Utilizes sophisticated system instructions to format AI outputs into highly actionable, clean Markdown tables and lists. Handles the transition from standard text prompts to multipart, vision-based payloads seamlessly.